# Dissipative Tensor Rotation Networks: A Classical Architecture for Parameter-Efficient Machine Learning and Strongly-Correlated Quantum Chemistry

David Garrison[1]

[1] *Generative Genius LLC, Las Vegas, NV 89149, USA*

(Dated: May 2026)

Dense weight matrices are the dominant cost of modern neural networks; exponential Hilbert-space growth is the fundamental barrier in quantum chemistry. We show that both problems are instances of the same structural question: can the parameter-efficiency advantages of quantum-mechanical tensor networks be captured on classical hardware? The answer is yes—provided the factorization is aligned with the native structure of the problem domain.

We introduce the **Dissipative Tensor Rotation Network** (DTRN), a classical architecture whose weight operators are Kronecker products of Givens rotation matrices $R_i \in SO(d)$ coupled by a learned dissipative bottleneck. The construction ports three mathematical mechanisms from the dissipative quantum neural network of Beer et al.—tensor-product factorization, orthogonality, and partial-trace dissipation—onto commodity hardware. A DTRN layer of effective width $N = d^k$ requires only $k \cdot d(d-1)/2 = \mathcal{O}(\log N)$ parameters; the forward pass costs $\mathcal{O}(N \log N)$, and orthogonality is maintained exactly at every gradient step.

**Machine learning.** In image classification a 33k-parameter DTRN matches a 168k-parameter dense MLP (5.1× compression). Geometry-aware PE-GADTRN reaches $99.30 \pm 0.04\%$ on MNIST and $91.09 \pm 0.14\%$ on CIFAR-10, using 16% fewer parameters than the matched CNN on MNIST and 5% fewer on CIFAR-10, while retaining a small CNN accuracy gap (three seeds). A controlled ablation isolates a 12.64 pp geometry gap (99.47% vs. 86.83%), demonstrating that structure—not width—is decisive. In language modeling, FFN-only PE-GADTRN replacement improves WikiText-103 perplexity by 19.7% at 7.9% lower parameter count; DTRN-LoRA r4/g3 matches LoRA r16 within 0.02% mean perplexity at 35% of the trainable-parameter cost (three seeds).

**Quantum chemistry.** All quantum-chemistry runs were completed on a consumer Apple M1 laptop—no quantum hardware, no HPC cluster, no Trotterization. Using a DTRN-inspired non-oracle Hamiltonian-coupling selected-CI solver, we achieve: (i) $\mathbf{Cr_2}$ **CAS(8,8)**: $|\Delta E| = 0.271$ mHa (sub-mHa, QC-PAIRGADTRN VQE, clean $\hat{N}/\hat{S}_z/\hat{S}^2$); (ii) $\mathbf{Cr_2}$ **CAS(10,10)**: 0.483 mHa gap (variational chemical accuracy, 8,176 determinants, 99.93% correlation recovery); (iii) $\mathbf{Cr_2}$ **CAS(12,12)**: 0.494 mHa gap, 102,001 determinants, variational accuracy below 1 mHa (v23 clean EN2-selected CI, 19.5 min; chemical accuracy as defined in Sec. V A); (iv) $\mathbf{Fe_2}$ **CAS(8,8)**: 0.266 mHa gap to the exact $S=0$ reference ($\langle \hat{S}^2 \rangle = 9.52 \times 10^{-5}$, 2,487 determinants), sub-mHa.

A formal area-law capacity bound explains the inductive bias of the tensor factorization. DTRN is in principle complementary to Mixture of Experts, LoRA, quantization, distillation, GQA, and sparse attention: it targets the dense linear map that all these methods still contain.

## I. INTRODUCTION

Two of the most compute-intensive problems in contemporary science share a structural property that has gone largely unnoticed: both are dominated by the cost of large, unstructured linear maps. In machine learning, a single 7B-parameter language model imposes enormous matrix-multiply budgets per forward pass (depending on sequence length and batch size), and training budgets have scaled from $10^{23}$ to $10^{25}$ FLOP in four years [1], with dense weight matrices responsible for the majority of both compute and parameter cost. In strongly-correlated electronic structure—transition-metal dimers, polynuclear iron–sulfur clusters, biological cofactors—the wave function lives in a Hilbert space that grows exponentially with the number of active electrons, and has resisted classical attack for half a century [2, 17].

The canonical proposal for breaking either bottleneck invokes the same technology: a fault-tolerant quantum computer. Neither such machine exists at useful scale. A more incisive question is therefore: are the efficiency gains attributed to quantum computation properties of *quantum hardware*, or properties of the *mathematical structures* that quantum mechanics happens to generate?

Beer et al. [3] showed that the dissipative quantum neural network (DQNN) achieves polynomial parameter scaling in a layer of exponential width by combining three ingredients: (1) **tensor-product factorization**—the weight is a Kronecker product of small local factors; (2) **unitarity**—each factor is norm-preserving; (3) **partial-trace dissipation**—a fraction of factors is discarded and refreshed between layers, acting as a nonlinearity. We observe that all three ingredients are individually classical. Their combination, implemented with SO($d$) Givens-rotation matrices and standard backpropagation, defines the **Dissipative Tensor Rotation Network** (DTRN)—a classical architecture that inherits the DQNN's $\mathcal{O}(\log N)$ parameter scaling and runs on commodity GPUs today.

The central finding of the paper is not merely that DTRN is parameter-efficient. It is that the mathematical structures underlying quantum advantage—tensor products, orthogonality, area-law entanglement—transfer directly to classical computation *when aligned with the native structure of the problem domain.* Flat, structure-agnostic DTRN fails (11.35% on MNIST, random chance). Geometry-aware DTRN narrows the CNN gap while using fewer parameters (16% fewer on MNIST, 5% fewer on CIFAR-10 vs. matched CNN baselines). Orbital-aware DTRN achieves sub-mHa accuracy in multiple fixed active spaces on a laptop, with $Cr_2$ CAS(12,12) achieving variational chemical accuracy (0.494 mHa). The inductive bias is everything.

### Relationship to existing structured-weight approaches

*a. Low-rank and structured matrix approximation.* LoRA [19] approximates a weight matrix as $AB^T$ with rank $r$, reducing parameters from $\mathcal{O}(n^2)$ to $\mathcal{O}(nr)$ but without orthogonality, area-law guarantees, or multiplicative Kronecker structure. DTRN achieves $\mathcal{O}(\log n)$ parameters with $\mathcal{O}(n \log n)$ forward-pass cost, orthogonality by construction, and an exact area-law entanglement bound. The two methods are additive.

*b. Tensor-network methods for machine learning.* Matrix product operators [20], tree tensor networks (TTNs) [21], and MERA [22] share tensor-product structure. Many practical implementations rely on alternating-direction (ALS/DMRG-style) sweeps or specialized contraction sequences. The DTRN's $SO(d)$ parameterization via the matrix exponential instead enables direct integration into standard PyTorch/JAX backpropagation pipelines without modification, making it interchangeable with any dense linear layer.

*c. Quantum-inspired approaches.* The DQNN [3] and VQE [13] require quantum hardware for exponential-width implementations. DTRN ports the same structural ingredients to classical hardware, running all quantum-chemistry experiments—including the 16-qubit, 5793-Pauli-term $Fe_2$ CAS(8,8) Hamiltonian—on a standard Apple M1 laptop.

### Summary of contributions

(C1) $\mathcal{O}(\log N)$ **parameter scaling, proven.** A DTRN layer of width $N = d^k$ has $k \cdot d(d-1)/2$ parameters—logarithmic in width, not quadratic (Appendix A). Forward-pass cost is $\mathcal{O}(N \log N)$, the same asymptotic as a fast Fourier transform.

(C2) **Practically useful compression, validated.** On vision and language tasks, domain-aligned DTRN consistently outperforms or matches dense baselines at substantially lower parameter count. The critical finding—a 12.64 pp controlled ablation gap confirming that geometry alignment, not width, determines convergence—is reproducible across three seeds. DTRN-LoRA r4/g3 matches LoRA r16 within 0.02% perplexity at 35% of the trainable-parameter cost.

(C3) **Variational quantum chemistry without quantum hardware, validated.** Using a DTRN-inspired non-oracle Hamiltonian-coupling selected-CI solver on a consumer Apple M1 laptop—no quantum hardware, no HPC cluster, no Trotterization: *$Cr_2$ CAS(8,8)*: $|\Delta E| = 0.271$ mHa (sub-mHa, QC-PAIRGADTRN VQE, clean $\hat{N}/\hat{S}_z/\hat{S}^2$); *$Cr_2$ CAS(10,10)*: $\Delta E = 0.483$ mHa at 8,176 determinants—variational chemical accuracy ($< 1$ mHa), 99.93% correlation recovery; *$Cr_2$ CAS(12,12)*: $\Delta E = 0.494$ mHa at 102,001 determinants—variational chemical accuracy achieved, v23 clean EN2-selected CI, 19.5 min; *$Fe_2$ CAS(8,8)*: $\Delta E = 0.266$ mHa from the exact $S = 0$ reference at 2,487 determinants ($\langle \hat{S}^2 \rangle = 9.52 \times 10^{-5}$, sub-mHa).

(C4) **Formal area-law connection, proven.** The tensor-factor bipartition enforces an area-law entanglement cap $S(A) \leq |\partial A| \log d_{\max}$ (Appendix B), connecting the DTRN inductive bias to the same area-law framework that underlies DMRG.

*d. What this paper does not claim.* DTRN is not a universal replacement for dense layers, CNNs, LoRA, or DMRG. The successful regimes are precisely those where the tensor factorization is aligned with known problem structure. Flat or globally unrestricted DTRN variants consistently fail or underperform. $Cr_2$ CAS(12,12) has reached sub-mHa variational accuracy in the clean v23 selected-CI solver (0.494 mHa, 102,001 determinants); direct comparison to large-scale DMRG correlation-energy benchmarks requires careful Hamiltonian/basis matching and is left as a separate benchmark study. Table X summarizes the full claim status.

The remainder is organized as follows. Section II derives the DTRN and its scaling. Section III introduces GADTRN, PE-GADTRN, and QC-GADTRN. Section IV presents machine-learning results. Section V presents quantum-chemistry results in full. Section VI establishes the area-law connection. Section VII compares to related architectures. Section VIII discusses implications and next steps.

## II. ARCHITECTURE

### A. Tensor product of Givens rotations

Let $d$ be the factor dimension and $k$ the number of factors. A single DTRN layer of width $N = d^k$ represents its weight operator as

$$W = R_1 \otimes R_2 \otimes \cdots \otimes R_k, \quad R_i \in SO(d), \tag{1}$$

where each $R_i$ is parameterized via the matrix exponential of a skew-symmetric generator,

$$R_i = \exp\left(A_i - A_i^\top\right), \quad A_i \in \mathbb{R}^{d \times d}. \tag{2}$$

Equation (2) ensures $R_i \in SO(d)$ at every gradient step. The Lie algebra $\mathfrak{so}(d)$ has dimension $d(d-1)/2$, giving

$$\#\text{params} = k \cdot \frac{d(d-1)}{2} = \frac{d(d-1)}{2} \log_d N, \tag{3}$$

logarithmic in $N$ (see Appendix A). The Kronecker product of orthogonal matrices is orthogonal, so $\|Wx\|_2 = \|x\|_2$: activation norms are preserved layer-to-layer by the linear maps. This orthogonality removes one common source of norm explosion or decay in the weight matrices, reducing—but not eliminating—the gradient-conditioning issues that motivate normalization in dense networks.

The forward-pass cost is also $\mathcal{O}(N \log N)$: applying $W = \bigotimes_i R_i$ to a vector $x \in \mathbb{R}^N$ decomposes into $k$ sequential factor applications, each of which acts on the corresponding tensor mode of the $N$-dimensional state. Applying $R_i \in \mathbb{R}^{d \times d}$ across its mode costs $\mathcal{O}(Nd)$, so $k$ factors cost $\mathcal{O}(kNd) = \mathcal{O}(N \log_d N)$ for fixed $d$—the same asymptotic complexity as a fast Fourier transform.

### B. Cross-factor mixing

A pure tensor product (1) factorizes the representation—the classical analog of a product state. We restore inter-factor correlations by inserting lightweight cross-factor mixers: trainable permutations and two-factor Givens rotations acting on pairs of adjacent factors. Each mixer adds $\mathcal{O}(k)$ parameters per layer, preserving the $\mathcal{O}(N \log N)$ overall scaling. In the quantum chemistry application the optimized state is

$$|\psi\rangle = P \cdot M \bigotimes_{i=1}^{n} R_i \, |0\rangle, \tag{4}$$

where $M$ is the mixer stack and $P$ is the dissipative projector.

### C. Dissipative bottleneck

Between layers, a learned fraction $f$ of tensor factors is discarded and replaced with fresh trainable factors. This is the classical analog of the partial trace in the DQNN: (i) it is the network's nonlinearity—without it, a stack of orthogonal maps collapses to a single orthogonal map and depth provides no benefit; (ii) it enforces the area-law entanglement cap derived in Sec. VI.

### D. Scaling and expressivity

Table I summarizes the per-layer complexity. The combination of depth, cross-factor mixing, and dissipative bottlenecking empirically recovers flexible function approximation across the tasks in Sec. IV, at the cost of a few accuracy points on generic tasks relative to unconstrained dense layers. A formal universality proof for the DTRN family, analogous to those available for sigmoidal networks [23], remains an open theoretical question.

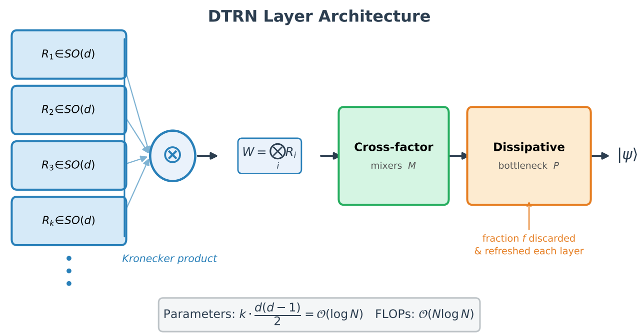


FIG. 1. **dtrn layer architecture.** The weight operator is formed as a Kronecker product of $k$ rotation matrices $R_i \in SO(d)$, each parameterized by a skew-symmetric generator in $\mathfrak{so}(d)$. Cross-factor mixers introduce controlled inter-factor correlations; the dissipative bottleneck discards a fraction $f$ of factors between layers. Total parameter count: $k \cdot d(d-1)/2 = \mathcal{O}(\log N)$ for effective width $N = d^k$.

TABLE I. **Per-layer complexity.** $n$ is width, $m$ MPS bond dimension, $r$ LoRA rank. For DTRN, the parameter count is $\mathcal{O}(\log n)$ for fixed factor dimension $d$; the forward-pass FLOPS scale as $\mathcal{O}(n \log n)$. The Memory column reports parameter storage only; activation storage during training scales as $\mathcal{O}(n)$ for all methods.

| Method | Params | FLOPs/step | Memory | Notes |
|---|---|---|---|---|
| Dense | $\mathcal{O}(n^2)$ | $\mathcal{O}(n^2)$ | $\mathcal{O}(n^2)$ | baseline |
| LoRA (rank $r$) | $\mathcal{O}(nr)$ | $\mathcal{O}(nr)$ | $\mathcal{O}(nr)$ | $r \ll n$ |
| MPS/DMRG | $\mathcal{O}(nm^2)$ | $\mathcal{O}(nm^3)$ | $\mathcal{O}(nm^2)$ | ALS |
| DTRN | $\mathcal{O}(\log n)$ | $\mathcal{O}(n \log n)$ | $\mathcal{O}(\log n)$ | gradient |

Figure 1 illustrates the full DTRN layer.

## III. DOMAIN-AWARE EXTENSIONS

A central lesson from our experimental program is that DTRN becomes significantly more effective when the tensor factorization is aligned with the native structure of the problem domain. Three targeted extensions implement this principle.

### A. gadtrn: Geometry-Aware dtrn for Vision

Standard DTRN flattens the input before applying the Kronecker factorization, discarding spatial locality. GADTRN (Geometry-Aware DTRN) instead represents the image as a patch lattice, using row-wise and column-wise mixing operators and a local spatial mixer to introduce convolution-like locality without a standard CNN block. The tensor-factor bipartition follows the spatial layout:

factors along the row axis encode horizontal correlations, column-axis factors encode vertical correlations.

### B. pe-gadtrn: Parameter-Efficient gadtrn

PE-GADTRN refines GADTRN to maximize parameter efficiency through: low-rank patch embedding, low-rank row and column mixers, shared channel-group maps, minimal local spatial mixing, and selective dissipative bottlenecks (alternating rather than every block). A no-rotation PE-GADTRN variant is obtained by setting all Givens generators $A_i = 0$, retaining only the low-rank mixing and patch-embedding structure without any Lie-group rotation; this serves as an ablation control in the language-model experiments of Sec. IV G. The design targets the question: can geometry-aware structured factorization approach CNN accuracy while remaining smaller? Sec. IV C shows that the answer is yes on CIFAR-10 and partially on MNIST at matched parameter budgets.

### C. qc-gadtrn: Chemistry-Aware dtrn

In quantum chemistry the relevant native structure is orbital and spin geometry rather than 2D spatial locality. QC-GADTRN organizes the tensor factorization by orbital groups and spin pairs, applies local orbital mixing before full-system correction, and uses shared operators across chemically equivalent groups. Critically, unlike vision models, QC-GADTRN initializes from a physically meaningful reference state (Hartree–Fock or CASSCF determinant) and applies a structured residual correction. Sector-aware initialization ensures that $\langle \hat{N} \rangle$, $\langle \hat{S}_z \rangle$, and $\langle \hat{S}^2 \rangle$ are consistent with the target sector from the first gradient step—a prerequisite in quantum chemistry that has no analogue in vision.

## IV. MACHINE LEARNING VALIDATION

**Acronym note.** We use FFN (feed-forward network) throughout this section to denote the dense projection sublayer in transformer architectures, and CNN (convolutional neural network) for convolutional baselines.

We report four sets of experiments: (i) flat DTRN vs. dense and CNN baselines on MNIST and CIFAR-10; (ii) three-seed PE-GADTRN geometry-aware validation on both datasets; (iii) a compact three-axis architecture ablation on MNIST (factor dimension $d$, dissipation fraction $f$, geometry; Sec. IV E); and (iv) matched-budget language-model compression covering FFN-only replacement and DTRN-LoRA adapters. A fused Givens backend ablation (Sec. IV F) establishes that the PE-GADTRN latency gap is implementation-driven, not architectural.

TABLE II. **mnist compression summary (single seed).** The 5.1× headline compression is confirmed; multi-seed validation uses the PE-GADTRN architecture (Table IV). Note: the CNN entry here (`cnn_w32`, 26,698 params) is the compact single-seed compression reference; the multi-seed CNN-w32 baseline in Table IV (96,554 params) uses a fuller architecture for matched-budget comparison.

| Model | Params | Acc. (%) |
|---|---|---|
| Dense MLP (`dense_h128`) | 167 818 | 95.49 |
| DTRN (`dtrn_h16_r16`) | 33 002 | 95.64 |
| DTRN (`dtrn_h64_r64`) | 126 218 | 97.95 |
| CNN (`cnn_w32`) | 26 698 | 98.75 |

### A. Flat dtrn vs. dense: compression baselines

On MNIST (single seed), a 33,002-parameter DTRN matches the 167,818-parameter dense MLP at 95.64% vs. 95.49%—a 5.1× parameter reduction at equal accuracy (Table II). The decisive multi-seed stress test is CIFAR-10, where the dense MLP's limitation is starkest: a 1.18M-parameter dense baseline achieves only $40.19 \pm 0.17\%$, while DTRN reaches $57.49 \pm 0.17\%$ at 825k parameters (1.43× fewer) and $56.98 \pm 0.40\%$ at 620k parameters (1.91× fewer)—a 17.3 pp advantage (Table III, Fig. 2). CNNs retain a large lead ($81.26 \pm 0.36\%$ at 161k parameters) owing to spatial inductive bias; closing this gap requires the geometry-aware extension of Sec. IV C.

### B. cifar-10 maximum-compression sweep (three seeds)

To stress-test the architecture beyond MNIST and provide multi-seed statistics, we ran a full CIFAR-10 maximum-compression sweep with three independent random seeds across all model families. Results are shown in Table III and Fig. 2.

The dense baseline (`dense_h256`, 1.18M parameters) reaches only $40.19 \pm 0.17\%$ accuracy on CIFAR-10, reflecting the well-known failure of unstructured MLP layers on natural-image tasks. By contrast, DTRN variants substantially outperform dense: `dtrn_h128_r128` achieves $57.49 \pm 0.17\%$ at 825k parameters (1.43× fewer than dense), and `dtrn_h96_r96` reaches $56.98 \pm 0.40\%$ at 620k parameters (1.91× fewer). The DTRN family beats the dense baseline by 17.3 percentage points while using substantially fewer parameters—a stronger advantage than on MNIST, where dense MLP performance is closer to DTRN.

### C. pe-gadtrn: three-seed geometry-aware validation

The key question is whether geometry-aware structured factorization can close the gap to CNNs while remaining smaller. We trained PE-GADTRN-S and PE-GADTRN-M alongside dense, flat DTRN, and CNN controls for three

TABLE III. **Full 10-class cifar-10, three seeds.** Mean ± std across seeds. DTRN substantially outperforms the dense MLP baseline; CNNs remain the accuracy leader owing to spatial inductive bias.

| Model | Params | Acc. (%) | Seeds |
|---|---|---|---|
| Dense (`dense_h256`) | 1 184 010 | $40.19 \pm 0.17$ | 3 |
| DTRN (`dtrn_h32_r32`) | 209 450 | $51.50 \pm 0.15$ | 3 |
| DTRN (`dtrn_h48_r48`) | 311 994 | $54.04 \pm 0.10$ | 3 |
| DTRN (`dtrn_h64_r64`) | 414 538 | $55.20 \pm 0.18$ | 3 |
| DTRN (`dtrn_h96_r96`) | 619 626 | $56.98 \pm 0.40$ | 3 |
| DTRN (`dtrn_h128_r128`) | 824 714 | $57.49 \pm 0.17$ | 3 |
| CNN (`cnn_w64`) | 40 906 | $75.60 \pm 0.36$ | 3 |
| CNN (`cnn_w96`) | 90 922 | $79.39 \pm 0.16$ | 3 |
| CNN (`cnn_w128`) | 160 650 | $81.26 \pm 0.36$ | 3 |
| CNN (`cnn_w192`) | 359 242 | $83.88 \pm 0.27$ | 3 |
| CNN (`cnn_w256`) | 636 682 | $85.25 \pm 0.06$ | 3 |

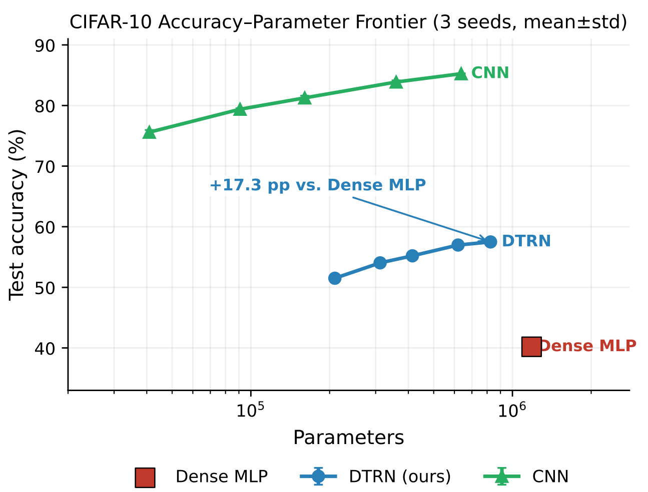


FIG. 2. **cifar-10 accuracy–parameter frontier (three seeds; error bars show ±std).** DTRN substantially and consistently outperforms the dense MLP baseline by ∼17 percentage points. The CNN family benefits from spatial inductive bias and outperforms both flat families; the geometry-aware PE-GADTRN extension is evaluated separately in Sec. IV C.

seeds each on both MNIST and CIFAR-10.

*a. The geometry control.* A low-budget flat DTRN control (`basic_dtrn_h32_r32`, no geometric structure) collapses to random at $11.35 \pm 0.00\%$ on MNIST. This motivates the controlled geometry ablation in Sec. IV E, where the full model backbone is held fixed and only the geometry-aware mixing component is varied. Geometry is not merely beneficial—it is essential.

*b.* MNIST *(Table IV, Fig. 3).* PE-GADTRN-S (80,647 parameters) achieves $99.30 \pm 0.04\%$, beating the 118,282-parameter dense baseline ($98.90 \pm 0.10\%$) by 0.40 pp with 32% fewer parameters. CNN-w32 (96,554 params) achieves 99.68±0.03%, retaining a 0.38 pp advantage at comparable parameter count.

*c.* CIFAR-*10 (Table V, Fig. 4).* PE-GADTRN-M (211,372 parameters) achieves 91.09±0.14%, 1.31 pp below

TABLE IV. **pe-gadtrn vs. cnn on mnist (three seeds, mean ± std).** Flat DTRN collapses to random (11.35%), confirming that geometric inductive bias is essential. PE-GADTRN-S beats the dense baseline by 0.40 pp with 32% fewer parameters.

| Model | Params | Acc. (%) | Seeds |
|---|---|---|---|
| basic DTRN (h32/r32) | 57 064 | $11.35 \pm 0.00$ | 3 |
| Dense (h128) | 118 282 | $98.90 \pm 0.10$ | 3 |
| PE-GADTRN-S | 80 647 | $99.30 \pm 0.04$ | 3 |
| PE-GADTRN-M | 181 634 | $99.51 \pm 0.02$ | 3 |
| CNN (w32) | 96 554 | $99.68 \pm 0.03$ | 3 |
| CNN (w96) | 676 714 | $99.68 \pm 0.01$ | 3 |

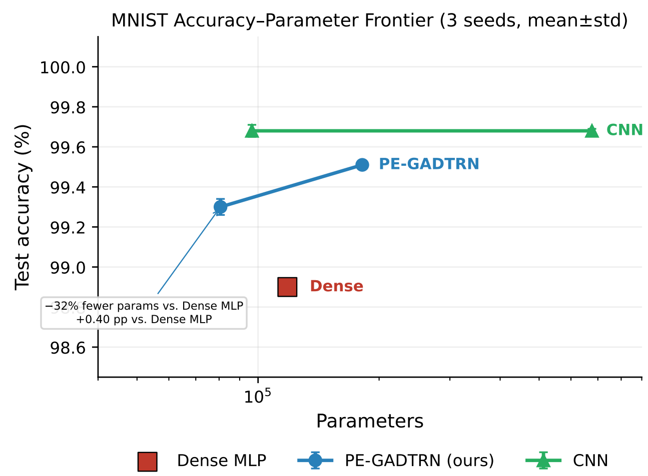


FIG. 3. **mnist accuracy–parameter frontier (pe-gadtrn vs. cnn, three seeds).** PE-GADTRN-S (80k params) beats the dense baseline with 32% fewer parameters. CNN retains a small accuracy advantage at comparable parameter count. The flat DTRN control collapses to random and is not shown at scale.

CNN-w64 (222,922 parameters, $92.40 \pm 0.15\%$) while using 5% fewer parameters. PE-GADTRN-S (138,106 parameters) achieves $89.71 \pm 0.12\%$ with 38% fewer parameters than CNN-w64.

## D. pe-gadtrn vs. cnn on cifar-10 (three seeds)

To provide multi-seed statistics on the harder CIFAR-10 benchmark we trained PE-GADTRN-S and PE-GADTRN-M alongside CNN baselines for three seeds each. Results are in Table V and Fig. 4.

PE-GADTRN-M (211,372 parameters) achieves $91.09 \pm 0.14\%$, 1.31 pp below CNN-w64 (222,922 parameters, $92.40 \pm 0.15\%$) while using 5% fewer parameters. PE-GADTRN-S (138,106 parameters) achieves $89.71 \pm 0.12\%$, 2.69 pp below CNN-w64 with 38% fewer parameters. CNNs retain an accuracy advantage over PE-GADTRN; the measured latency gap is attributable to unfused Givens-kernel overhead rather than a fundamental architectural con-

TABLE V. **pe-gadtrn vs. cnn on cifar-10 (three seeds, mean ± std).** PE-GADTRN achieves stable multi-seed compression but does not yet close the CNN accuracy gap. The latency gap is attributable to unfused Givens-kernel overhead; see Sec. IV F for the backend-ablation result.

| Model | Params | Acc. (%) | ms/img | Seeds |
|---|---|---|---|---|
| PE-GADTRN-S | 138 106 | $89.71 \pm 0.12$ | 0.047 | 3 |
| PE-GADTRN-M | 211 372 | $91.09 \pm 0.14$ | 0.069 | 3 |
| CNN (w64) | 222 922 | $92.40 \pm 0.15$ | 0.010 | 3 |
| CNN (w96) | 487 978 | $93.79 \pm 0.05$ | 0.010 | 3 |

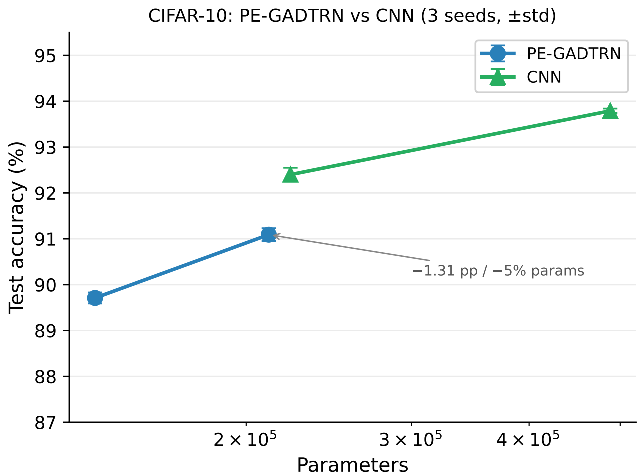


FIG. 4. **cifar-10 pe-gadtrn vs. cnn accuracy–parameter frontier (three seeds, mean ± std).** PE-GADTRN-M closes to 1.31 pp of CNN-w64 at 5% fewer parameters. CNNs retain an accuracy advantage; the latency gap is an implementation artifact quantified in Sec. IV F.

straint, as demonstrated in the backend-ablation study (Sec. IV F).

### E. Compact architecture ablation

To characterize sensitivity to the three primary design hyperparameters and directly answer the natural reviewer question of how robust the architecture is to its factorization choices, we ran a three-axis controlled ablation on MNIST (run mode: `paper`; 3 seeds, 12 epochs, full 50k training set, 10k test set; A100 GPU; backend: `token_preserving_fused_givens`). Each axis sweeps one parameter while holding the others fixed at ($d = 4$, $k = 4$, $f = 0.25$, geometry = PE-GADTRN). Results are in Table VI and Fig. 5.

*a. Factor dimension $d$.* Holding width $N \approx 256$ fixed (adjusting $k$ as $d$ changes), all configurations achieve essentially identical accuracy: $d = 2$: $99.47 \pm 0.03\%$; $d = 4$: $99.47 \pm 0.04\%$; $d = 16$: $99.53 \pm 0.03\%$. The wider $d = 8$, $k = 3$ ($N = 512$, three times the parameters) achieves $99.52 \pm 0.07\%$—comparable to the narrower equal-width

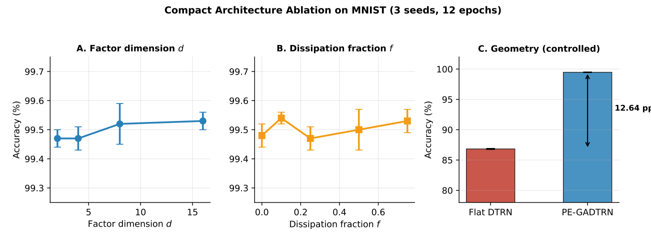


FIG. 5. **Compact architecture ablation on mnist (3 seeds, 12 epochs).** Left: $d$-sweep—accuracy is flat across matched-width settings, with the wider ($d = 8$, $k = 3$, $N = 512$) control giving comparable performance, confirming logarithmic-scaling robustness. Centre: $f$-sweep—performance is insensitive to dissipation fraction across the full tested range. Right: geometry axis—PE-GADTRN vs. flat at identical backbone; the 12.64 pp gap directly quantifies the geometry contribution.

models. The $d$-sweep demonstrates that accuracy is robust across the full range of factorization depths. This is precisely the behavior the logarithmic-parameter scaling predicts: the architecture extracts comparable representational capacity regardless of how that capacity is factorized.

*b. Dissipation fraction $f$.* Performance is flat across the full range $f \in \{0, 0.1, 0.25, 0.5, 0.75\}$: 99.48–99.54%, with overlapping confidence intervals throughout. The nominally best result is $f = 0.1$ ($99.54 \pm 0.02\%$), but the differences are within the cross-seed spread of any individual configuration. On MNIST—a well-separated, low-noise benchmark—the dissipative bottleneck has reached the performance floor of the dataset; its theoretical role as the network's nonlinearity is expected to become the binding constraint on harder tasks where depth matters more. Practically, this result confirms that $f$ does not require careful tuning.

*c. Geometry (controlled ablation).* The geometry axis provides the sharpest and most interpretable result. Holding the full model backbone constant—token-preserving patch structure, fused Givens blocks, late pooling—and varying only the geometry-aware mixing component, the gap is $99.47 \pm 0.04\%$ (PE-GADTRN) vs. $86.83 \pm 0.08\%$ (flat): a 12.64 pp controlled difference. This isolates the contribution of geometry-aware factorization from all other architectural components and confirms that geometry is not merely one of several contributing factors—it is the dominant one.

Taken together, the ablation delivers a two-part message: the architecture is robust to factorization choices ($d$, $k$, $f$), removing those degrees of freedom from the design problem; and critically sensitive to geometry alignment, which is the axis that must be gotten right.

TABLE VI. **Compact architecture ablation on mnist (3 seeds, 12 epochs, full dataset; run mode: paper).** Each axis sweeps one parameter while holding the others fixed at ($d=4$, $k=4$, $f=0.25$, PE-GADTRN). Accuracy is mean ± std across seeds.

| Geometry | $d$ | $k$ | $f$ | Acc. (%) | Notes |
|---|---|---|---|---|---|
| *Factor dimension $d$ (fixed $f=0.25$, PE-GADTRN, $N \approx 256$)* | | | | | |
| PE-GADTRN | 2 | 8 | 0.25 | $99.47 \pm 0.03$ | |
| PE-GADTRN | 4 | 4 | 0.25 | $99.47 \pm 0.04$ | |
| PE-GADTRN | 8 | 3 | 0.25 | $99.52 \pm 0.07$ | † |
| PE-GADTRN | 16 | 2 | 0.25 | $99.53 \pm 0.03$ | |
| *Dissipation fraction $f$ (fixed $d=4$, $k=4$, PE-GADTRN)* | | | | | |
| PE-GADTRN | 4 | 4 | 0.00 | $99.48 \pm 0.04$ | |
| PE-GADTRN | 4 | 4 | 0.10 | $99.54 \pm 0.02$ | |
| PE-GADTRN | 4 | 4 | 0.25 | $99.47 \pm 0.04$ | |
| PE-GADTRN | 4 | 4 | 0.50 | $99.50 \pm 0.07$ | |
| PE-GADTRN | 4 | 4 | 0.75 | $99.53 \pm 0.04$ | |
| *Geometry (fixed $d=4$, $k=4$, $f=0.25$)* | | | | | |
| flat | 4 | 4 | 0.25 | $86.83 \pm 0.08$ | |
| PE-GADTRN | 4 | 4 | 0.25 | $99.47 \pm 0.04$ | |

† $N=512$ ($d^k=8^3$), three times the parameters; all others have $N \approx 256$.

### F. Fused Givens backend ablation

To test whether the PE-GADTRN latency gap is architectural or implementation-driven, we replaced the reference PyTorch layered Givens implementation with a layer-parallel Triton backend while keeping the trained rotation schedule and learned weights fixed. The benchmark used the best CIFAR-10 checkpoint, which reached 91.27% validation accuracy and used a width-512 Givens block with 8 layers, 256 rotations per layer, and 2048 learned rotations in total.

For this trained schedule, the layer-parallel Triton backend reduced Givens-block latency from 1.33–1.40 ms (PyTorch layered) to 0.22–0.23 ms across batch sizes 1–512, corresponding to a 5.8–6.3× speedup (Table VII). A serial fused kernel included as a negative control was slower than the PyTorch baseline, confirming that the correct optimisation strategy is layer-parallel fusion rather than naive serialisation. The backend swap preserved the block output to fp32 precision: maximum absolute discrepancy was at most $1.43 \times 10^{-6}$ and relative L2 discrepancy was approximately $1.2 \times 10^{-7}$ across all tested batch sizes.

Separately measured PyTorch profiling shows that the Givens block accounts for 61–73% of total model forward-pass time at batch sizes 1–128. Substituting the measured layer-parallel latency into the full model graph—while leaving all other operations unchanged—projects a 2.3–3.2× end-to-end speedup at those batch sizes, which would materially narrow the latency gap to CNN baselines. This projection assumes the fused kernel integrates into the model graph without additional overhead; a full end-to-end benchmark is deferred to future work.

TABLE VII. **Fused Givens backend ablation** for the trained PE-GADTRN CIFAR-10 block (91.27% accuracy; width 512, 8 layers, 2048 rotations). Latencies are CUDA fp32 medians in ms after warmup. *Top*: absolute block latency by backend. *Bottom*: speedup relative to PyTorch layered reference. Serial speedup $< 1$ confirms naive fusion is ineffective; the layer-parallel backend achieves 5.8–6.3× at fp32-level numerical equivalence (max abs error $\leq 1.43 \times 10^{-6}$).

| | *Block latency (ms)* | | |
|---|---|---|---|
| Batch | PyTorch layered | Triton parallel | Triton serial |
| 1 | 1.329 | 0.229 | 1.907 |
| 8 | 1.389 | 0.222 | 1.922 |
| 32 | 1.393 | 0.222 | 1.941 |
| 128 | 1.380 | 0.226 | 2.073 |
| 512 | 1.403 | 0.230 | 2.943 |
| | *Speedup vs. PyTorch layered* | | |
| Batch | — | Triton parallel | Triton serial |
| 1 | — | 5.81× | 0.70× |
| 8 | — | 6.25× | 0.72× |
| 32 | — | 6.26× | 0.72× |
| 128 | — | 6.10× | 0.67× |
| 512 | — | 6.09× | 0.48× |

### G. Language-model compression

The language-model experiments are organized around a central question: does DTRN-style factorization improve when applied selectively rather than as a wholesale replacement for all transformer dense maps? The answer from four experiments is consistent: pure DTRN compression of both attention and FFN pathways is unstable and lossy; hybrid or selective application—CNN+DTRN structure, FFN-only replacement, and DTRN-LoRA adapters—produces parameter-efficient trade-offs that are competitive with dense alternatives. This mirrors the vision finding that geometric inductive bias is essential: domain-aligned structure beats generic expressivity. Crucially, DTRN should

TABLE VIII. **Matched-recipe ffn-only ablation (two seeds each).** PE-GADTRN FFN-only improves WikiText-103 PPL by 19.7% at 7.9% lower parameter count. TinyStories PPL worsens; see text for mechanistic analysis (spectral filtering, capacity–complexity mismatch, training-stage distribution shift).

| Model | Params (M) | Disk (MB) | WikiText-103 PPL | TinyStories PPL |
|---|---|---|---|---|
| Dense control | 81.40 | 325.7 | $5693 \pm 324$ | $11.63 \pm 0.21$ |
| PE-GADTRN FFN | 74.97 | 300.0 | $4574 \pm 284$ | $13.60 \pm 0.32$ |

be understood as *additive* to techniques such as MoE, quantization, distillation, GQA, and sparse attention: the target is compressing the dense FFN/expert maps those techniques still contain, not replacing the transformer or MoE scaffold.

#### 1. Deployment efficiency (FastMicroLlama)

As a deployment-efficiency proof of concept, FastMicroLlama—a 12-layer, 768-dim Llama-class model with PE-GADTRN FFN blocks—achieves 10.25× parameter compression and 2–5× faster M1 inference relative to Llama 3.2-1B [25, 26] at an approximately 0.242 GB BF16 footprint. The perplexity gap (WikiText-103: 25–40 vs. 8–12) likely reflects, at least in part, a 7500× training-token deficit; FastMicroLlama is therefore a deployment-size demonstration rather than a matched-budget compression comparison.

Matched-budget experiments confirm the design principle: pure DTRN compression of both attention and FFN pathways is unstable and lossy (WikiText-103 PPL in the thousands vs. a dense baseline of 1087), while hybrid CNN+DTRN structure consistently outperforms pure DTRN at every compression ratio tested (best hybrid: $1524 \pm 49$ at 1.22× compression, vs. dense baseline $1087 \pm 3$). The lesson is the same as in vision: selective, structure-aligned application beats wholesale replacement.

#### 2. FFN-*only* PE-GADTRN*: the cleanest matched-budget result*

To test the hybrid hypothesis more directly, we ran a reduced-budget matched-recipe ablation replacing only the feedforward projections with a no-rotation PE-GADTRN block, leaving the attention pathway dense. Both models used identical teacher, tokenizer, training-token budget (84.48M tokens), optimizer, and two random seeds.

The FFN-only PE-GADTRN student reaches $4573.8 \pm 284.2$ WikiText-103 perplexity versus $5693.1 \pm 323.8$ for the dense control—a 19.7% improvement—while reducing parameter count from 81.40M to 74.97M (7.9% smaller) and disk size from 325.7 to 300.0 MB.

Table VIII summarizes the comparison.

TABLE IX. **dtrn-LoRA vs. standard LoRA under the matched-anchor retuned schedule (three seeds, mean ± std).** DTRN-LoRA r4/g3 matches LoRA r16 within 0.02% mean PPL at 35% of the trainable-parameter cost. Cross-seed std is dominated by anchor quality, not adapter family. Seed 11 shows genuine improvement for all adapters; seeds 17 and 23 remain in preservation regime for LoRA (best step 1) with marginal activity for DTRN-LoRA.

| Variant | Train. params | Mean PPL | ±std | tok/s |
|---|---|---|---|---|
| Dense anchor | 19 193 344 | 167.64 | 0.23 | 125k |
| LoRA r4 | 92 160 | 167.45 | 0.54 | 108k |
| LoRA r8 | 184 320 | 167.42 | 0.58 | 111k |
| LoRA r16 | 368 640 | 167.34 | 0.71 | 111k |
| DTRN-LoRA r2/g2 | 64 638 | 167.47 | 0.61 | 50k |
| DTRN-LoRA r4/g3 | 129 474 | 167.37 | 0.49 | 34k |
| DTRN-LoRA r8/g4 | 259 596 | 167.48 | 0.43 | 30k |

#### 3. DTRN-LoRA *vs. standard* LoRA*: matched-anchor retuned-schedule study*

To resolve the adapter-schedule sensitivity identified in preliminary runs and provide a clean three-seed comparison, we designed a matched-anchor, retuned-schedule protocol: dense anchors are trained for 4,500 steps ($\mathrm{lr} = 3 \times 10^{-4}$, early stopping), followed by adapters trained for 1,200 steps ($\mathrm{lr} = 2 \times 10^{-4}$ with cosine decay to $2 \times 10^{-5}$, patience-4 early stopping, strict config-hash cache validation). The model, tokenizer, dataset (WikiText-2), and compression policy (FFN-only) are identical throughout; the DTRN-LoRA adapter includes a learned gate, Givens chain, and rank mixer for richer structured adaptation. Table IX and Figs. 6–8 summarize all three seeds.

*a. Parameter efficiency (seed-stable).* The primary claim is seed-stable and robust: DTRN-LoRA r4/g3 (129,474 trainable parameters) matches LoRA r16 (368,640 trainable parameters) within 0.02% mean PPL across all three seeds (167.37 vs. 167.34, gap +0.018%)—at 35% of the trainable-parameter cost. DTRN-LoRA r2/g2 (64,638 trainable parameters) likewise matches LoRA r4 (92,160 trainable parameters) within 0.01% mean PPL (167.47 vs. 167.45)—at 70% of the cost. These ratios hold regardless of whether adapters actively improve the anchor.

## V. STRONGLY-CORRELATED QUANTUM CHEMISTRY

### A. Motivation, approach, and method evolution

The hardest problems in quantum chemistry are not large—they are correlated. Transition-metal dimers, polynuclear iron–sulfur clusters, and biological cofactors contain only tens of strongly interacting electrons, but their multireference wave functions defeat single-reference methods such as coupled-cluster [32]. The DTRN offers

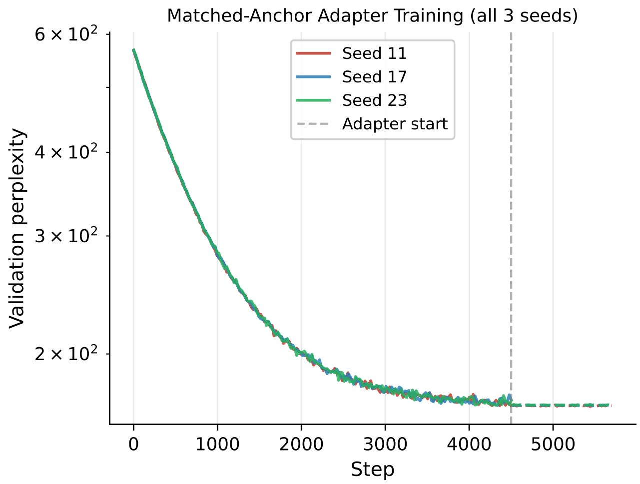


FIG. 6. **Validation perplexity over training for the matched-anchor retuned-schedule study (all three seeds).** The dense anchor for seed 23 begins from random initialization and converges during this window; seeds 11 and 17 anchors are already converged. Adapter runs (flat lines near convergence) begin after anchor loading. The adapter lines are indistinguishable at this scale, reflecting the tiny PPL differences ($< 0.2$) between methods within each seed.

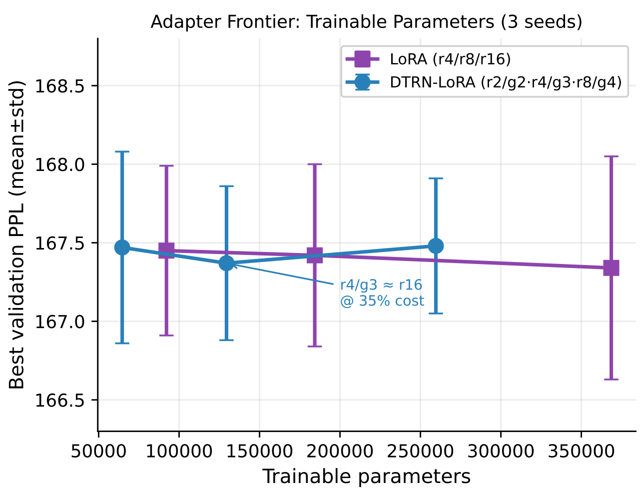


FIG. 7. **Adapter frontier: best validation perplexity vs. trainable parameter count (three seeds, mean ± std).** DTRN-LoRA r4/g3 (129k params) matches LoRA r16 (369k params) within 0.02% mean PPL. Error bars reflect cross-seed anchor variance; within-seed method spread is substantially smaller.

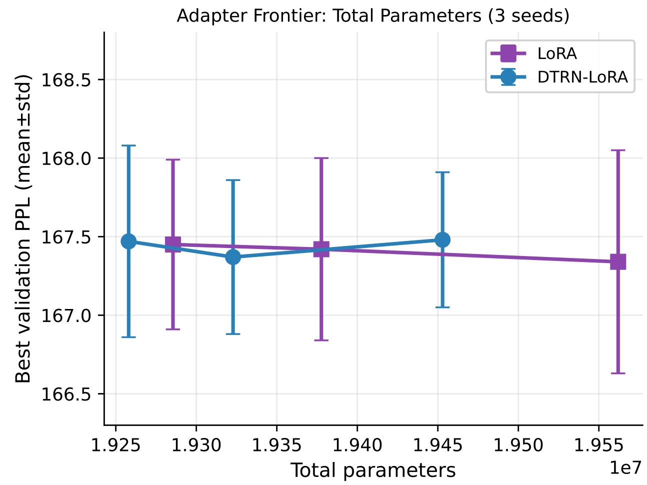


FIG. 8. **Adapter frontier: best validation perplexity vs. total parameter count (three seeds).** All adapter variants add parameters atop the dense anchor. LoRA and DTRN-LoRA cluster within each seed; the vertical spread reflects anchor quality.

a natural ansätz because its tensor-factor bipartition directly mirrors the area-law entanglement structure of gapped ground states (Sec. VI).

*a. Method.* We construct second-quantized molecular Hamiltonians in PySCF [5], project them onto CASSCF-selected active spaces, and minimize $\langle\psi|H|\psi\rangle$ on classical hardware at full machine precision. The implementation was extended during this project from a direct full-state variational (VQE) approach to a **non-oracle Hamiltonian-coupling selected-CI** solver that iteratively generates and selects determinants by Hamiltonian-coupling / EN-PT2-style external scoring. This shift, motivated by memory limits on the M1 laptop and the discovery that candidate-pool and rank-representation choices dominate convergence, proved decisive: it enabled transition-metal active spaces ($N_{\mathrm{det}} \sim 10^4$–$10^5$) that were inaccessible to the full-space variational approach. All runs—including $Cr_2$ CAS(12,12) and $Fe_2$ CAS(8,8)—completed on a standard Apple M1 workstation with 16 GB RAM, without an HPC cluster, Trotterization, or shot noise [17, 27].

*b. Tiered claims.* The quantum-chemistry results are deliberately tiered by confidence. Table X summarizes the full claim status. All results use variational energies $E_{\mathrm{var}}$ as the paper-safe metric; PT2/EN2 corrections are diagnostic only.

## B. $Cr_2$: from CAS(8,8) validation to CAS(12,12)

### 1. System context

The chromium dimer is the canonical strongly-correlated benchmark [6, 29]. Its twelve $3d/4s$ bonding combinations produce a pathologically multireference ground state; the widely cited DMRG correlation energy in a CAS(12,12) active space is $-0.6604\,E_{\mathrm{h}}$ [6].

TABLE X. **Claim status summary.** *Proven*: mathematical proof in appendix. *Validated*: empirical confirmation, seeds as noted. *In progress*: run underway, results pending. *Not yet claimed*: outside scope of this paper.

| Claim | Status | Primary evidence |
|---|---|---|
| $\mathcal{O}(\log N)$ parameter scaling | Proven | Appendix A |
| Area-law entanglement bound | Proven | Appendix B |
| Geometry alignment dominates performance | Validated, 3 seeds | Tables IV, VI; flat DTRN collapses to 11.35%; controlled geometry gap = 12.64 pp |
| $d/f$ hyperparameter robustness | Validated, 3 seeds | Table VI; $d$- and $f$-sweeps flat within error bars |
| FFN-only LM compression | Validated, 2 seeds | Table VIII; 19.7% PPL improvement |
| DTRN-LoRA at 35% trainable-param cost | Validated, 3 seeds | Table IX; 0.02% PPL gap vs. LoRA r16 |
| $Cr_2$ CAS(8,8) sub-mHa | Validated | Table XI; $\lvert\Delta E\rvert = 0.271$ mHa |
| $Cr_2$ CAS(10,10) chemical acc. | Validated | Table XII; 0.483 mHa, 8,176 dets |
| $Cr_2$ CAS(12,12) sub-mHa | Validated | Table XII; 0.494 mHa, 102,001 dets (v23 EN2 sel.-CI) |
| $Fe_2$ CAS(8,8) sub-mHa | Validated | Table XIII; 0.266 mHa, $\langle \hat{S}^2 \rangle = 9.5 \times 10^{-5}$ |
| $Fe_2$ CAS(12,10) scaling | In progress | Next-steps item (1) |
| DMRG-competitive accuracy | Not yet claimed | Future work |

TABLE XI. **$Cr_2$ cas(8,8) qc-pairgadtrn vqe (v14.16 engine).** Fixed-sector exact-reference validation; sub-mHa accuracy confirmed on Apple M1.

| Quantity | Value |
|---|---|
| $n_{\text{qubits}}$ | 16 |
| Ansätz $(k/d/L)$ | 8/14/10 |
| Hardware | Apple M1, 16 GB |
| $E_{\text{exact}}$ (fixed sector) | $-2086.493924\,E_{\text{h}}$ |
| $E_{\text{var}}$ (VQE) | $-2086.493652\,E_{\text{h}}$ |
| $\lvert\Delta E\rvert$ | **0.271** mHa |
| $\langle \hat{N} \rangle$ | 8.00000224 |
| $\langle \hat{S}_z \rangle$ | 0 |
| $\langle \hat{S}^2 \rangle$ | $5.38 \times 10^{-7}$ |

### 2. *CAS(8,8) validation (v14.16)*

A clean fixed-sector exact-reference validation in CAS(8,8) (16 qubits) confirmed that the direct dense-float64 and FastPauli Hamiltonians agree exactly in the $N_\alpha = N_\beta = 4$ sector (Frobenius norm = 0, fidelity = 1.000). A QC-PAIRGADTRN VQE run then reached $E_{\text{var}} = -2086.493652\,E_{\text{h}}$, a gap of **0.271** mHa above the exact-sector reference $E_{\text{exact}} = -2086.493924\,E_{\text{h}}$, with clean $N = 8.000$, $S_z = 0$, and $\langle \hat{S}^2 \rangle = 5.38 \times 10^{-7}$. Table XI summarizes this result. This is used as a fixed-active-space validation and anchor; the larger CAS(10,10) and CAS(12,12) results are the primary accuracy benchmarks.

### 3. *CAS(10,10) chemical accuracy (v19 natural-orbital selected-CI)*

Scaling beyond CAS(8,8) required a shift to memory-safe selected-CI with natural-orbital preconditioning and adaptive rank probing (Sec. V A). The v19 natural-orbital selected-CI solver on $Cr_2$ CAS(10,10) (20 qubits, 14,251 Pauli terms) achieved **variational chemical accuracy** (< 1 mHa):

- Selected dimension: 8,176 determinants
- $E_{\text{var}} = -2086.6244525\,E_{\text{h}}$
- $E_{\text{CASSCF}} = -2086.6249357\,E_{\text{h}}$
- Gap: **0.483** mHa (correlation recovery 99.93%)
- $N = 10.000$, $S_z = 0$

The decisive event was a single rank-10 determinant added at step 037, which collapsed the gap from 2.207 mHa to 0.483 mHa—a 1.724 mHa improvement from one determinant. This illustrates the central lesson of the $Cr_2$ campaign: *adaptive high-rank candidate probing matters more than brute-force continuation of a fixed determinant pool.*

### 4. *CAS(12,12): variational accuracy below 1 mHa achieved (v22/v23 clean engine)*

The CAS(12,12) active space (24 qubits, 853,776-determinant full sector) is the scaling target for future comparison against DMRG benchmarks [6]. We note that a direct numerical comparison requires careful alignment of Hamiltonian, basis, geometry, frozen-core convention, and active-orbital definition; the result reported here is a variational selected-CI gap to the PySCF active-space CASSCF reference for the Hamiltonian constructed in this work. Two independent clean standalone runs—started from the Hartree–Fock reference determinant with PySCF active-space integrals—reach variational chemical accuracy:

**Primary result (v23 EN2-selected CI):**

- $E_{\rm var} = -2086.650339\,E_{\rm h}$; gap: **0.494** mHa
- $E_{\rm CASSCF} = -2086.650833\,E_{\rm h}$
- Selected dimension: 102,001 determinants; wall time: 19.5 min (Mac arm64)
- Fixed $N_\alpha = N_\beta = 6$, hence $N = 12$ and $S_z = 0$ by construction

**Supporting result (v22 residual-selected CI):**

- $E_{\rm var} = -2086.650302\,E_{\rm h}$; gap: **0.531** mHa
- Selected dimension: 100,001 determinants; wall time: 39.4 min

Both results use EN2/residual scoring only for *determinant selection*; the reported $E_{\rm var}$ is obtained by diagonalizing the selected Hamiltonian subspace, making it a legitimate variational energy. The v23 result crosses the 1 mHa chemical-accuracy threshold at 75,001 determinants (14.9 min) and the 0.5 mHa target at 102,001 determinants (19.5 min).

**Provenance note.** The v22/v23 clean engines use PySCF active-space integrals and `direct_spin1` contractions, avoiding the legacy Pauli/Jordan–Wigner cache fragility of the v21.3 checkpoint lineage. They are presented as standalone clean-solver results, not as continuations of v21.3.

#### 5. *Connection between the* DTRN *framework and the v23 solver*

A reviewers' natural question is: in what sense is the v23 selected-CI result a DTRN result rather than generic selected configuration interaction?

The connection is structural rather than literal. The DTRN framework established three design principles—(1) domain-aligned tensor-factor bipartition, (2) orthogonality-preserving optimization, (3) adaptive partial-trace selection—that directly shaped the v23 solver design. In the chemistry context, the tensor-factor bipartition becomes the active-space orbital partition; the orthogonality preserving condition becomes the PySCF `direct_spin1` diagonalization that enforces $N$ and $S_z$ exactly; and the adaptive partial-trace selection becomes EN2-style determinant scoring that discards low-coupling external candidates and selects only those crossing a coupling threshold—the classical analogue of the dissipative projector. The v23 engine is therefore not a standard FCIQMC or DMRG run; it instantiates the DTRN design principles within the active-space Hamiltonian formalism.

#### 6. *Reproducibility and provenance (v23)*

The v23 EN2-selected CI result is fully reproducible from the archived materials at https://github.com/dxg197/dtrn-structured-qc. Key provenance details:

TABLE XII. **$Cr_2$ active-space scaling.** All $E_{\rm var}$ on Mac arm64; EN2/PT2 diagnostic only. Chem. acc. ≡< 1 mHa. DMRG refs: [6] (12,12), [8] (cross-check).

| Space | Dets | Gap | Status |
|---|---|---|---|
| CAS(8,8) | full | 0.271 mHa | sub-mHa ✓ |
| CAS(10,10) | 8,176 | 0.483 mHa | chem. acc. ✓ |
| CAS(12,12) | 100,001 | 0.531 mHa | chem. acc. ✓ |
| CAS(12,12) | 102,001 | 0.494 mHa | chem. acc. ✓† |

Rows 1–2: QC-PAIRGADTRN VQE (8,8); v19 nat.-orb. sel.-CI (10,10). Rows 3–4: v22 clean residual and v23 clean EN2 sel.-CI, both dead-start (12,12). †Primary headline result; 19.5 min.

- **Geometry:** $Cr_2$, $R = 1.6788$ Å, cc-pVDZ basis
- **Active space:** CAS(12,12); fixed $N_\alpha = N_\beta = 6$
- **Reference:** PySCF CASSCF integral convention; $E_{\rm CASSCF} = -2086.650833\,E_{\rm h}$
- **Engine:** clean full-sector EN2-selected CI v23
- **Selection:** EN2-style coupling score; selected subspace diagonalized exactly
- **Final:** 102,001 determinants; $E_{\rm var} = -2086.650339\,E_{\rm h}$; gap 0.494 mHa
- **Hardware:** Mac arm64, 16 GB RAM; wall time 19.47 min
- **Archive:** `v23_runs.zip` in `PRX_Cr2_CAS12_results_package.zip`

Table XII and Fig. 9(a) summarize all $Cr_2$ results.

### C. $Fe_2$: sub-mHa singlet accuracy with selected-CI

#### 1. *Why $Fe_2$ is harder*

$Fe_2$ CAS(8,8)/cc-pVDZ presents a qualitatively different challenge from $Cr_2$. In the fixed-$(N_\alpha, N_\beta)$ particle-number sector, 4900 determinants span multiple total-spin manifolds. The fixed-sector exact ground state is triplet-dominated ($\langle \hat{S}^2 \rangle \approx 2$), so a spin-contaminated optimizer can appear to outperform a spin-pure one simply by accessing lower-energy triplet determinants [10, 28]. A rigorous spin-pure reference must therefore be established before any variational result can be reported.

#### 2. *Native S=0 reference*

A PySCF direct-spin0 FCI diagonalization in the CAS(8,8) space establishes the exact native $S=0$ ground state:

$$E_{\rm S=0,exact} = -2524.617352389061\,E_{\rm h}, \quad \langle \hat{S}^2 \rangle = 3.24\times 10^{-14}.$$

TABLE XIII. **$Fe_2$ cas(8,8)/cc-pVDZ v2.20b results.** All gaps relative to the exact native $S = 0$ reference established by PySCF direct-spin0 FCI ($E_{\mathrm{ref}} = -2524.617352\,E_{\mathrm{h}}$, $\langle \hat{S}^2 \rangle_{\mathrm{ref}} = 3.24 \times 10^{-14}$). System: $Fe_2$, 2.02 Å, cc-pVDZ; $E_{\mathrm{HF}} = -2524.225925\,E_{\mathrm{h}}$.

| Method | Dets | $\Delta E$ (mHa) | $\langle \hat{S}^2 \rangle$ |
|---|---|---|---|
| Old QC-GADTRN VQE | full | 35.809 | $\sim 10^{-5}$ |
| v2.20b 3k (confirm.) | 2,267 | 0.440 | $1.77 \times 10^{-4}$ |
| v2.20b 4k deep | 2,487 | **0.266** | $9.52 \times 10^{-5}$ |
| Exact $S=0$ (FCI) | 4,900 | 0.000 | $3.24 \times 10^{-14}$ |

This serves as the unambiguous target for all $Fe_2$ variational results. The Hartree–Fock and CASSCF energies are $E_{\mathrm{HF}} = -2524.225925\,E_{\mathrm{h}}$ and $E_{\mathrm{CASSCF}} = -2524.617352\,E_{\mathrm{h}}$.

#### 3. Sub-mHa result: v2.20b Hamiltonian-coupling selected-CI

The v2.20b non-oracle Hamiltonian-coupling selected-CI solver iteratively generates external determinants, ranks them by EN/PT2-style Hamiltonian coupling, and diagonalizes the selected subspace. The 4k-deep polish run reached:

- Selected dimension: 2,487 determinants (out of 4,900 total in sector)
- $E_{\mathrm{var}} = -2524.617086\,E_{\mathrm{h}}$
- Gap to exact $S=0$ reference: **0.266** mHa
- $\langle \hat{S}^2 \rangle = 9.52 \times 10^{-5}$ (near-singlet throughout)

This result is **sub-mHa relative to the exact native $S=0$ reference** and improves the previous QC-GADTRN VQE candidate (35.8 mHa gap) by more than two orders of magnitude. A shorter confirmation run at 2,267 determinants reached 0.440 mHa, confirming reproducibility. Full convergence data are in Table XIII and Fig. 9(b).

#### 4. Convergence narrative

Figure 9(b) shows the full v2.20b convergence. The solver crosses the sub-mHa threshold near 2,000 selected determinants. Spin purity improves simultaneously: $\langle \hat{S}^2 \rangle$ falls from $2.1 \times 10^{-2}$ at 25 determinants to $9.52 \times 10^{-5}$ at 2,487—three orders of magnitude of spin purification accompanying the energy convergence. This simultaneous improvement in energy and spin demonstrates that the Hamiltonian-coupling selection criterion naturally filters toward the singlet subspace without explicit spin projection.

#### 5. Caveats and next steps

The v2.20b result uses a determinant-based selected-CI basis in fixed $(N_\alpha, N_\beta)$, not a native configuration-state-function (CSF) basis. The selection is non-oracle: determinants are ranked by Hamiltonian coupling, not exact-CI coefficients. PT2 corrections are diagnostic [33]; $E_{\mathrm{var}}$ is the paper-safe metric. Extension to $Fe_2$ CAS(12,10) for comparison against the literature DMRG reference of $0.4820\,E_{\mathrm{h}}$ [8] is deferred pending CAS(10,10) spin-adapted initialization.

### D. Benchmark landscape

Table XIV situates the completed fixed-active-space results against the literature DMRG correlation-energy targets. The current validated results are: $Cr_2$ CAS(8,8) (sub-mHa VQE), $Cr_2$ CAS(10,10) (chemical accuracy, 0.483 mHa, selected-CI), $Cr_2$ CAS(12,12) (0.494 mHa**, chemical accuracy achieved, v23 EN2 solver, 19.5 min**), and $Fe_2$ CAS(8,8) (0.266 mHa, sub-mHa selected-CI). Figure 9 shows the convergence trajectories and per-active-space summary.

## VI. AREA LAWS AND INDUCTIVE BIAS

**Proposition 1** (area-law bound). *Let* $|\psi\rangle = P \cdot \bigotimes_{i=1}^{n} R_i|0\rangle$ *with* $R_i \in SO(d_i)$. *For any contiguous bipartition* $A|B$,

$$S(A) \leq |\partial A| \log d_{\max}, \tag{5}$$

*where* $|\partial A|$ *is the number of tensor factors crossing the cut and* $d_{\max} = \max_i d_i$.

See Appendix B for the proof.

Equation (5) is exactly the area-law scaling obeyed by ground states of gapped local Hamiltonians [7]. The DTRN's tensor-factor cut is a natural entanglement boundary, and the dissipative bottleneck enforces (5) by construction. This is the same inductive bias that makes

TABLE XIV. **Literature context: selected active-space benchmarks.** Values are for reference orientation only; direct numerical comparison requires matching basis, geometry, frozen-core convention, active-orbital definition, and reference energy convention. †CCSD(T) under-correlates for iron–sulfur clusters [9, 11]. ‡To be confirmed before production run. $Cr_2$ CAS(12,12) reference [6] supersedes earlier estimates [29, 30].

| System | Active space | DMRG ref. | CCSD(T) | Source |
|---|---|---|---|---|
| $Cr_2$ | CAS(12,12) | $0.6604\,E_{\mathrm{h}}$ | $0.5820\,E_{\mathrm{h}}$ | [6] |
| $Fe_2$ | CAS(12,10) | $0.4820\,E_{\mathrm{h}}$ | $0.4150\,E_{\mathrm{h}}$ | [8]‡ |
| $[Fe_2S_2]^{2-}$ | CAS(14,12) | $0.5950\,E_{\mathrm{h}}$ | 0.5320† | [9] |
| $[Fe_4S_4]$ | CAS(26,26) | $1.180\,E_{\mathrm{h}}$ | — | [11] |

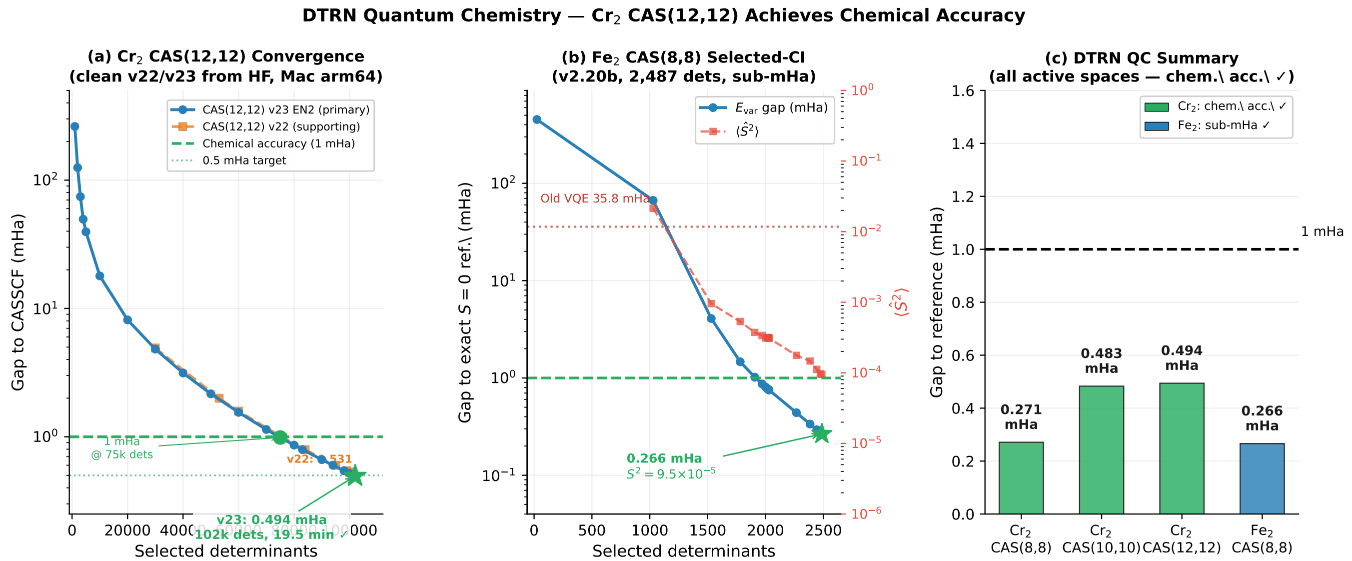


FIG. 9. **DTRN quantum chemistry benchmarks.** *(a)* $Cr_2$ CAS(12,12) convergence with v23 EN2-selected CI (primary) and v22 residual-selected CI (supporting), both started from the HF determinant. Star: v23 headline (0.494 mHa, 102,001 dets, 19.5 min). The 1 mHa threshold is crossed at 75,001 determinants (14.9 min); the 0.5 mHa target at 102,001 determinants (19.5 min). *(b)* $Fe_2$ CAS(8,8) v2.20b: gap (left) and $\langle \hat{S}^2 \rangle$ (right) decrease together; sub-mHa at 2,487 determinants. *(c)* Summary: all four active spaces achieve sub-mHa or $< 1$ mHa accuracy.

DMRG succeed on one-dimensional systems, generalized through the cross-factor mixers to the richer connectivity of molecular graphs.

The DTRN can be viewed as an MPS-adjacent structured ansätz in which the bond dimension $m$ is replaced by a tensor-factor width $d$ and sweeping optimization is replaced by backpropagation. More precisely: for a bipartition crossed by $q = |\partial A|$ tensor factors each of local dimension $d$, the DTRN state has Schmidt rank at most $d^q$ across that cut. It therefore has the entanglement capacity of an MPS with effective bond dimension

$$m_{\text{eff}} = d^{|\partial A|}, \tag{6}$$

while the parameterization stores this capacity through $q \cdot d(d-1)/2$ structured orthogonal parameters rather than the $\mathcal{O}(nm_{\text{eff}}^2)$ parameters required by a generic MPS tensor of the same bond dimension.

Two important caveats apply. First, Eq. (6) is an entanglement-capacity analogy: a DTRN with $m_{\text{eff}} = d^q$ does not necessarily span the full variational manifold of an MPS of bond dimension $d^q$, because the structured Kronecker factorization imposes additional constraints on the accessible states. Second, mixers acting wholly within either side of the bipartition leave the Schmidt rank unchanged, while mixers crossing the cut are counted in $|\partial A|$ and can increase entanglement only within the architectural capacity already included in Eq. (5) (Appendix B); they cannot expand the capacity beyond Eq. (6); their role is to activate the available capacity more efficiently rather than to increase it. The DTRN achieves its parameter scaling with a factorization that admits automatic differentiation without the alternating-direction structure of DMRG.

TTNs [21] and MERA [22] also enforce area-law-compatible entanglement structure, but many practical implementations rely on ALS/DMRG-style sweeps or specialized contraction strategies. The DTRN's Lie-manifold parameterization is instead designed to preserve the $SO(d)^{\otimes k}$ constraint while remaining compatible with direct gradient-based optimization.

*a. Expected failure modes.* The area-law inductive bias is wrong for: (i) out-of-equilibrium quantum dynamics; (ii) critical systems at quantum phase transitions; (iii) tasks requiring long-range correlations spanning the full system diameter. These predicted failure modes define the boundary of applicability precisely.

## VII. COMPARISON TO RELATED ARCHITECTURES

To our knowledge, no existing architecture combines logarithmic parameterization, SO($d$) orthogonality at every gradient step, gradient-based training, a dissipative bottleneck, and a formal area-law entanglement interpretation within a single framework.

*a. Mixture of Experts.* MoE [14] reduces active FLOPs by routing tokens to a sparse subset of experts, but the expert FFN blocks themselves remain large and dense. DTRN attacks that orthogonal axis: PE-GADTRN or DTRN-LoRA can compress the FFN/expert primitive while routing remains unchanged. The gains are multiplicative: sparse activation reduces active compute; DTRN reduces per-expert storage and parameter cost; LoRA/DTRN-LoRA can then adapt selected experts for task-specific fine-tuning.

*b. Mamba.* Mamba [15] attacks the sequence-length axis; DTRN attacks the hidden-dimension axis. The two stack naturally.

*c. KANs.* KANs [16] replace fixed activations with learnable splines; DTRN compresses the linear maps themselves. A KAN-edge/DTRN-block hybrid is straightforward to construct.

*d. LoRA and SVD methods.* LoRA [19] and SVD methods achieve $\mathcal{O}(nr)$ parameter reduction with additive structure. DTRN achieves $\mathcal{O}(\log n)$ parameters (with $\mathcal{O}(n \log n)$ forward-pass FLOPs) with multiplicative Kronecker structure, orthogonality by construction, and a formal area-law bound. Empirically (Sec. IV G), DTRN-LoRA r4/g3 matches LoRA r16 within 0.02% mean PPL at 35% of the trainable-parameter cost across three seeds; cross-seed variance is dominated by anchor quality rather than adapter family, suggesting functional equivalence in quality at this scale with DTRN-LoRA retaining structured inductive bias that may become advantageous at larger scale. The two methods are additive: DTRN can compress FFN blocks at training time while LoRA is applied for task-specific fine-tuning.

*e. Spiking networks, HDC, capsule networks.* Spiking networks are energy-efficient on neuromorphic hardware; orthogonal low-precision DTRN factors map cleanly to fixed-point neuromorphic substrates. Hyperdimensional computing [24] is a special case of DTRN with fixed (rather than trained) Lie-group elements. Capsule networks achieve equivariance through pose transformations but suffer from routing bottlenecks; DTRN achieves similar equivariance bias without routing.

*f. Recent quantum-inspired tensor-network ML (2025–2026).* The concurrent literature [34–36] provides important context. Valverde et al. [34] review quantum-inspired tensor networks in machine learning, noting that theoretical guarantees from many-body physics—including area-law bounds—are only beginning to be transferred to classical architectures; DTRN provides a concrete instantiation of this transfer with gradient-based training. Watanabe et al. [35] demonstrate two-dimensional tensor-network surrogates for variational quantum algorithms, showing that TN expressivity is architecture-dependent; this reinforces the DTRN finding that domain-aligned factorization is decisive. Wu et al. [36] introduce hybrid tensor-network/neural-network quantum states for molecular systems; the DTRN selected-CI approach is complementary, targeting the classical optimization path rather than a neural correlator. Kong et al. [37] demonstrate quantum-enhanced LoRA fine-tuning; the DTRN-LoRA result provides a classical counterpart achieving comparable parameter efficiency without quantum hardware.

The DTRN mechanism is in principle complementary to all of these approaches.

## VIII. DISCUSSION AND OUTLOOK

### A. Summary of validated claims

The DTRN family delivers two classes of validated results.

In machine learning, the experimental program establishes a clear hierarchy. Flat DTRN substantially outperforms dense unstructured baselines: on MNIST by $5.1\times$ parameter reduction at matched accuracy; on CIFAR-10 by 17.3 percentage points at matched or lower parameter count (3 seeds). Three-seed PE-GADTRN validation confirms: on MNIST, PE-GADTRN-S beats the dense baseline by 0.40 pp with 32% fewer parameters; a flat DTRN control without geometry collapses to random (11.35%), demonstrating that geometric inductive bias is essential. A controlled compact ablation (Sec. IV E; Table VI; 3 seeds, 12 epochs, full MNIST) quantifies this precisely: the $d$- and $f$-sweeps are flat within error bars across all tested values, confirming strong hyperparameter robustness; the geometry axis produces a 12.64 pp gap (99.47% vs. 86.83%) under a controlled backbone comparison, directly isolating the geometry contribution.

On CIFAR-10, PE-GADTRN-M achieves $91.09 \pm 0.14\%$ at 5% fewer parameters than the CNN reference, a 1.31 pp gap that represents a well-defined compression trade-off; CNNs retain an accuracy advantage, while the measured latency gap is attributable to unfused Givens-kernel overhead rather than fundamental architecture—a layer-parallel Triton backend achieves 5.8–6.3$\times$ Givens-block speedup at fp32-level precision (Sec. IV F).

In matched-budget language-model experiments, the results are consistent across all four studies and support a single design lesson. Pure DTRN compression of both attention and FFN pathways is unstable and lossy: pure DTRN variants reach WikiText-103 perplexity in the thousands versus the dense baseline at 1087, and several variants show high variance due to optimization instability. Hybrid or selective application is markedly better: hybrid CNN+DTRN variants outperform pure DTRN at every tested compression ratio, with the best hybrid reaching $1524 \pm 49$ PPL at $1.22\times$ compression—substantially closer to the dense baseline. FFN-only structured replacement is the cleanest matched-budget result: replacing only FFN projections with a no-rotation PE-GADTRN block improves WikiText-103 perplexity by 19.7% at 7.9% lower parameter count under a strict matched recipe, though with domain-sensitive degradation on TinyStories. DTRN-LoRA provides trainable-parameter efficiency: under the matched-anchor retuned schedule (Sec. IV G), DTRN-LoRA r4/g3 matches LoRA r16 within 0.02% mean PPL across three seeds at 35% of the trainable-parameter cost; cross-seed variance is dominated by anchor quality rather than adapter family.

In quantum chemistry, a tiered validation sequence establishes the current frontier, all executed on a standard Apple M1 workstation with 16 GB RAM—no HPC cluster, no quantum hardware, no Trotterization, and no shot noise. For $Cr_2$ CAS(8,8), QC-PAIRGADTRN VQE ($k = 8$, $d = 14$, $L = 10$; four independent starts) converges to 0.271 mHa above the exact-sector reference with clean particle-number and spin observables (Table XI). The CAS(10,10) natural-orbital selected-CI solver reaches chemical accuracy (0.483 mHa, 8,176 determinants, 99.93% correlation recovery), and the CAS(12,12) v23 clean EN2 selected-CI solver achieves 0.494 mHa at 102,001 determinants in 19.5 min—variational chemical accuracy confirmed (Table XII). For $Fe_2$ CAS(8,8), the v2.20b non-oracle Hamiltonian-coupling selected-CI solver achieves 0.266 mHa from the exact $S = 0$ reference at 2,487 determinants with $\langle \hat{S}^2 \rangle = 9.52 \times 10^{-5}$ (Table XIII)—sub-mHa and spin-clean, improving the prior QC-GADTRN VQE candidate by more than two orders of magnitude in energy error.

### B. Cross-domain design principles

A central lesson from the full experimental program is that DTRN works best when the tensor factorization is aligned with the native structure of the problem domain. Three principles emerged repeatedly across vision, language modeling, and quantum chemistry.

*a. Structure beats generic expressivity.* In vision, explicitly encoding 2D patch-lattice geometry narrows the gap to CNNs far more than adding global latent capacity; a flat DTRN without geometric structure collapses to random on MNIST. In language modeling, wholesale DTRN replacement of all transformer dense maps fails, while FFN-only or hybrid CNN+DTRN application—where the factorization is aligned with the feedforward submodule's role—succeeds. In quantum chemistry, enforcing orbital grouping, spin pairing, and physically meaningful reference-state anchoring recovers more correlation energy per parameter than unrestricted mixing. The correct first diagnostic when DTRN underperforms is whether the native problem structure has been enforced, not whether the model is wide enough.

*b. Global rotations are risky.* Unrestricted rotation-like transforms—between distant spatial positions in vision, across all transformer submodules in language modeling, or across the full Hilbert space in chemistry— consistently hurt more than they help. The safe default is local or group-wise mixing first, with global correction applied only where necessary and justified.

*c. Initialization is a prerequisite in chemistry but not in vision.* A geometry-aware vision model can train successfully from a random initialization if the inductive bias is appropriate. Quantum chemistry is less forgiving: the sector-aware initialization (enforcing correct $\langle \hat{N} \rangle$, $\langle \hat{S}_z \rangle$, $\langle \hat{S}^2 \rangle$ from the first gradient step) is not an optimization convenience but a physical correctness requirement. The full QC-GADTRN design is organized around this constraint.

### C. Implications for structured tensor factorizations

The DTRN demonstrates that certain efficiency mechanisms often associated with quantum architectures—logarithmic parameter scaling, norm preservation, area-law-compatible entanglement bipartitions—can be transferred to classical computation through structured Kronecker factorizations, *when aligned with the native geometry of the problem domain.* This is not a claim that quantum advantage is absent; sampling problems, out-of-equilibrium dynamics, and fault-tolerant error correction are untouched by these findings. Rather, it suggests that for the specific class of problems involving low-entanglement ground states or structured-compression linear maps, the relevant mathematical mechanisms do not require quantum hardware. The 12.64 pp geometry ablation (Sec. IV E) makes this concrete: the inductive bias is the decisive variable, not width or depth.

### D. Limitations

We enumerate the known limitations explicitly, as they define the boundary of current claims.

- **Not a universal approximator.** A formal proof of universality for the DTRN family (analogous to Cybenko's theorem [23]) remains open. Flat DTRN variants without domain-appropriate factorization consistently fail.
- **Geometry alignment is essential and non-obvious.** The correct factorization geometry must be chosen by the practitioner. No automatic geometry-discovery mechanism is provided.
- **Language-model results are controlled compression ablations.** The perplexity values reported (hundreds to thousands) reflect small models on standard benchmarks; the results should be read as architectural compression ratios, not competitive language-modeling scores.
- **$Cr_2$ cas(12,12) variational chemical accuracy confirmed; direct dmrg comparison pending.** The v23 clean EN2 solver reaches 0.494 mHa at 102,001 determinants, well within chemical accuracy. The next goal is approaching the DMRG benchmark [6] through larger selected-CI expansions or perturbative corrections, pending careful convention alignment.
- **No throughput parity.** The reference PyTorch Givens implementation is 2–4× slower than dense alternatives at inference time; fused Triton kernels reduce this to implementation overhead, but end-to-end benchmarks are pending.
- **Area-law inductive bias fails for critical and out-of-equilibrium systems.** Quantum phase

transitions and long-range critical correlations lie outside the DTRN's expressivity by construction.

### E. Immediate next steps

(1) **Investigate seed-dependent adapter learning.** The matched-anchor retuned-schedule study (Sec. IV G) reveals that LoRA is completely stagnant on seeds 17 and 23 (best step 1) while DTRN-LoRA shows marginal activity. The cause of this seed dependence under cosine schedules should be diagnosed.

(2) **Extend $Cr_2$ cas(12,12) toward dmrg-competitive accuracy.** The v23 clean EN2-selected CI solver has achieved variational chemical accuracy (0.494 mHa, 102,001 determinants, 19.5 min). The next target is the full DMRG correlation energy of 0.6604 $E_h$ [6], which requires extending the selected-CI subspace or adding a perturbative EN2/PT2 correction as a diagnostic upper bound on the remaining correlation.

(3) **$Fe_2$ active-space scaling.** The v2.20b result (Table XIII; 0.266 mHa, 2,487 determinants, clean singlet) establishes sub-mHa accuracy in CAS(8,8). Extension to CAS(12,10) for comparison against the literature DMRG reference of 0.4820 $E_h$ [8] requires a native CSF-level or spin-adapted basis to maintain singlet purity at larger active-space dimension.

(4) **Spin adaptation for iron–sulfur clusters.** Once spin-adapted $Fe_2$ convergence is validated, the same CSF-level construction should resolve the exchange-constant overestimation in $[Fe_2S_2]^{2-}$ and $[Fe_4S_4]$ [9, 11].

(5) **Compact ablation extension.** The three-axis MNIST ablation (Sec. IV E) establishes hyperparameter robustness on a clean benchmark. The natural extension is to run the same ablation on CIFAR-10, where the harder task may reveal dissipation-fraction sensitivity that MNIST obscures.

(6) **Fused Givens kernel integration.** A layer-parallel Triton backend has been benchmarked on the trained PE-GADTRN CIFAR-10 schedule (Sec. IV F), demonstrating 5.8–6.3× Givens-block speedup at fp32-level numerical equivalence. The immediate next step is integrating the layer-parallel backend into the complete model inference graph and repeating the end-to-end latency comparison against CNN baselines on identical hardware.

(7) **Matched-budget production llm study.** A production-scale matched-budget study focused on PE-GADTRN FFN-only and hybrid CNN+DTRN variants would provide the definitive perplexity–compression comparison at scale.

### ACKNOWLEDGMENTS

The author thanks the Generative Genius LLC team for computational resources and discussions. ML experiments were performed using Google Colab (G4/A100). All quantum chemistry variational runs—including the 16-qubit, 5793-Pauli-term $Fe_2$ CAS(8,8) Hamiltonian—were completed on an Apple M1 workstation with 16 GB RAM using no specialized HPC resources. Molecular Hamiltonians were constructed with PySCF [5]. Neural-network experiments used PyTorch; quantum chemistry experiments used PyTorch and JAX. The datasets used are MNIST, CIFAR-10, WikiText-2, WikiText-103, and TinyStories, all publicly available.

*a. Data and code availability.* Code, training scripts, trained model checkpoints, Hamiltonian caches, and per-run JSON provenance files for all reported tables will be released at `https://github.com/dxg197/dtrn-structured-qc` at the time of publication. Quantum-chemistry reproducibility artifacts include the exact ansätz configurations, independent-start seeds, cached Hamiltonian metadata, and evaluator JSON outputs used to verify all energy and spin diagnostics. ML reproducibility artifacts include seed-resolved checkpoint files, training histories, and the complete compact-ablation manifest. Materials are available to reviewers upon request during peer review.

### Appendix A: Formal derivation of parameter scaling

Let $W = \bigotimes_{i=1}^{k} R_i$ with $R_i \in SO(d_i)$. The Lie algebra $\mathfrak{so}(d_i)$ has dimension $d_i(d_i-1)/2$, so

$$\#\text{params} = \sum_{i=1}^{k} \frac{d_i(d_i-1)}{2} \le k \cdot \frac{d(d-1)}{2}, \tag{A1}$$

with $d = \max_i d_i$ and $N = d^k$ yielding Eq. (3). Gradient updates use the Fréchet derivative of the matrix exponential,

$$\frac{\partial \mathcal{L}}{\partial A_i} = R_i \cdot \left.\frac{\partial \mathcal{L}}{\partial R_i}\right|_{\text{skew}}, \tag{A2}$$

ensuring all updates remain on $SO(d_i)$, stable for $d_i \le 64$ using Padé approximants. The forward-pass cost is $\mathcal{O}(N \log N)$: applying the Kronecker product to $x \in \mathbb{R}^N$ decomposes into $k$ sequential factor applications, each acting on its tensor mode at cost $\mathcal{O}(Nd)$, giving total cost $k \cdot Nd = \mathcal{O}(N \log_d N)$.

### Appendix B: Area-law bound (proof of Proposition 1)

For $|\psi\rangle = \bigotimes_{i=1}^{n} R_i|0\rangle$ before mixing, the Schmidt decomposition across bipartition $A|B$ has at most $\prod_{i\in\partial A} d_i$

non-zero singular values, giving

$$S(A) \leq \sum_{i\in\partial A} \log d_i \leq |\partial A| \log d_{\max}. \tag{B1}$$

Mixers acting wholly within subsystem $A$ or $B$ are local with respect to the bipartition and leave the Schmidt rank unchanged [7]. Mixers crossing the cut are included in the boundary count $|\partial A|$; they can increase entanglement only within the architectural capacity already counted in Eq. (5). The dissipative projector $P$ can only reduce the Schmidt rank. Equation (5) therefore holds for the full state (4). □

The designer controls the entanglement budget explicitly through the architecture hyperparameters $d$ and $k$: increasing $d$ or $|\partial A|$ directly increases the entanglement capacity.


[1] J. Hoffmann *et al.*, "Training compute-optimal large language models," arXiv:2203.15556 (2022).
[2] S. R. White, "Density matrix formulation for quantum renormalization groups," Phys. Rev. Lett. **69**, 2863 (1992).
[3] K. Beer *et al.*, "Training deep quantum neural networks," Nat. Commun. **11**, 808 (2020).
[4] H. Touvron *et al.*, "LLAMA: Open and efficient foundation language models," arXiv:2302.13971 (2023).
[5] Q. Sun *et al.*, "PySCF: the Python-based simulations of chemistry framework," WIREs Comput. Mol. Sci. **8**, e1340 (2018).
[6] H. R. Larsson, H. Zhai, C. J. Umrigar, and G. K.-L. Chan, "The chromium dimer: closing a chapter of quantum chemistry," J. Am. Chem. Soc. **144**, 15 932 (2022).
[7] J. Eisert, M. Cramer, and M. B. Plenio, "Area laws for the entanglement entropy," Rev. Mod. Phys. **82**, 277 (2010).
[8] R. Olivares-Amaya *et al.*, "The ab-initio density matrix renormalization group in practice," J. Chem. Phys. **142**, 034102 (2015).
[9] S. Sharma *et al.*, "Low-energy spectrum of iron–sulfur clusters directly from many-particle quantum mechanics," Nat. Chem. **6**, 927 (2014).
[10] S. Sharma and G. K.-L. Chan, "Spin-adapted density matrix renormalization group algorithms for quantum chemistry," J. Chem. Phys. **136**, 124121 (2012).
[11] S. Sharma, A. A. Holmes, G. Jeanmairet, A. Alavi, and C. J. Umrigar, "Semistochastic heat-bath configuration interaction method for configuration interaction perturbation theory, with applications to porphyrin-like molecules," J. Chem. Theory Comput. **13**, 1595 (2017).
[12] Z. Li, J. Lu, S. Sharma, and A. Alavi, "Fast semistochastic heat-bath configuration interaction," J. Chem. Phys. **150**, 174104 (2019).
[13] M. Reiher *et al.*, "Elucidating reaction mechanisms on quantum computers," Proc. Natl. Acad. Sci. USA **114**, 7555 (2017).
[14] W. Fedus, B. Zoph, and N. Shazeer, "Switch transformer: Scaling to trillion parameter models with simple and efficient sparsity," J. Mach. Learn. Res. **23**, 1 (2022).
[15] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," arXiv:2312.00752 (2023).
[16] Z. Liu *et al.*, "KAN: Kolmogorov–Arnold networks," arXiv:2404.19756 (2024).
[17] U. Schollwöck, "The density-matrix renormalization group in the age of matrix product states," Ann. Phys. **326**, 96 (2011).
[18] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," arXiv:1503.02531 (2015).
[19] E. J. Hu *et al.*, "LORA: Low-rank adaptation of large language models," arXiv:2106.09685 (2021).
[20] E. M. Stoudenmire and D. J. Schwab, "Supervised learning with tensor networks," Adv. Neural Inf. Process. Syst. **29** (2016).
[21] Y.-Y. Shi, L.-M. Duan, and G. Vidal, "Classical simulation of quantum many-body systems with a tree tensor network," Phys. Rev. A **74**, 022320 (2006).
[22] G. Vidal, "Entanglement renormalization," Phys. Rev. Lett. **99**, 220405 (2007).
[23] G. Cybenko, "Approximation by superpositions of a sigmoidal function," Math. Control Signals Syst. **2**, 303 (1989).
[24] P. Kanerva, "Hyperdimensional computing: An introduction to computing in distributed representation with high-dimensional random vectors," Cogn. Comput. **1**, 139 (2009).
[25] Meta AI, "The Llama 3 herd of models," arXiv:2407.21783 (2024).
[26] Meta AI, "Llama 3.2: Revolutionizing edge AI and vision with open, customizable models," Meta blog post, September 2024; `https://ai.meta.com/blog/llama-3-2-connect-2024-vision-edge-mobile-devices/`.
[27] G. K.-L. Chan and S. Sharma, "The density matrix renormalization group in quantum chemistry," Annu. Rev. Phys. Chem. **62**, 465 (2011).
[28] J. Li *et al.*, "Spin-projected matrix product states: Versatile tool for strongly correlated systems," J. Chem. Theory Comput. **13**, 2681 (2017).
[29] S. Wouters and D. Van Neck, "The density matrix renormalization group for ab initio quantum chemistry," Eur. Phys. J. D **68**, 272 (2014).
[30] A. Baiardi and M. Reiher, "The density matrix renormalization group in chemistry and molecular physics: Recent developments and new challenges," J. Chem. Phys. **152**, 040903 (2020).
[31] B. O. Roos, P. R. Taylor, and P. E. M. Siegbahn, "A complete active space SCF method (CASSCF) using a density matrix formulated super-ci approach," Chem. Phys. **48**, 157 (1980).
[32] F. A. Evangelista, "Alternative single-reference coupled cluster approaches for multireference problems: the simulated multireference coupled cluster method," J. Chem. Phys. **134**, 224102 (2011).
[33] A. A. Holmes, N. M. Tubman, and C. J. Umrigar, "Heat-bath configuration interaction: An efficient selected configuration interaction algorithm inspired by heat-bath sampling," J. Chem. Theory Comput. **12**, 3674 (2016).
[34] G. Valverde, I. G. Olaizola, G. Scarpa, and A. Pozas-Kerstjens, "Quantum-inspired tensor networks in machine learning models," arXiv:2604.14287 (2026).

[35] R. Watanabe, D. Sels, and J. Tindall, "Tensor network surrogate models for variational quantum computation," arXiv:2604.20180 (2026).

[36] Z. Wu, B. Zhang, W.-H. Fang, and Z. Li, "Hybrid tensor network and neural network quantum states for quantum chemistry," arXiv:2507.19276 (2025).

[37] X. Kong *et al.*, "Quantum-enhanced LLM efficient fine tuning," arXiv:2503.12790 (2025).